Computer Color Systems

Lecture 3

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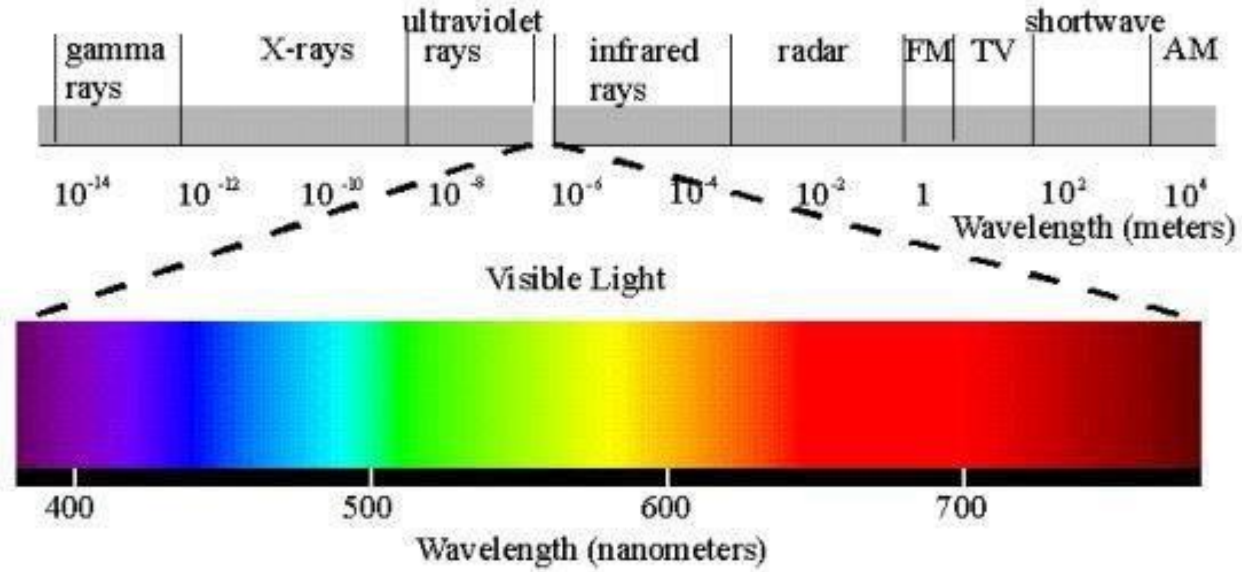
Outline

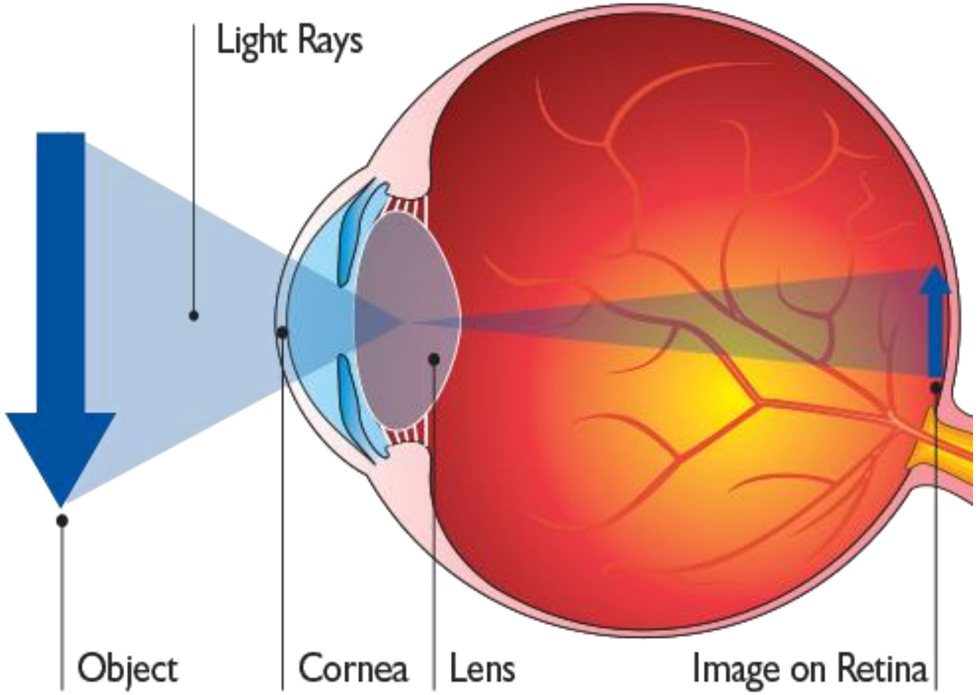
- How human see colors?
- Three-color theory
- Additive and Subtractive Color
- Color Models and how they are used in graphics.
  - RGB
  - CMY(K)
  - HSV

Light

- Light is the part of the electromagnetic spectrum that causes a reaction in our visual systems.
- Generally these are wavelengths in the range of about 350-750 nm (nanometers)
- Long wavelengths appear as reds and short wavelengths as blues



Human vision



- Our eyes work by focusing light through an elastic lens, onto a patch at the back of our eye called the retina.
- The retina contains light sensitive rod and cone cells that are sensitive to light, and send electrical impulses to our brain that we interpret as a visual stimulus.

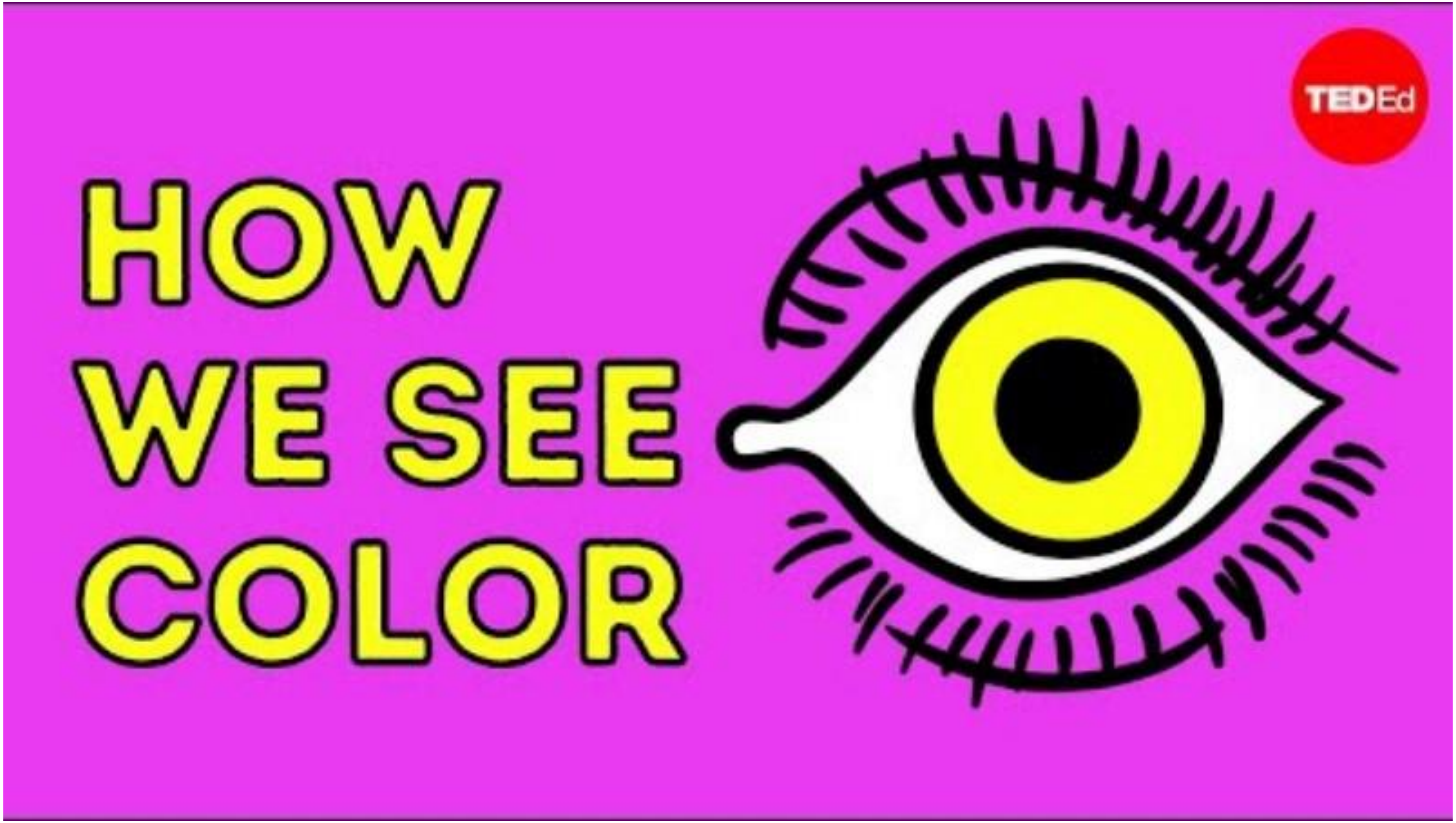
Three-Color Theory(trichromatic theory)

“human eyes only perceive three colors of light: red, blue, and green. The wavelengths of these three colors can be combined to create every color on the visible light spectrum. -Young-Helmholtz ”

- Human visual system has two types of sensors
  - Rods: monochromatic, night vision
  - Cones
    - Color sensitive
    - Three types of cone
    - Only three values (the tristimulus values) are sent to the brain
- Need only match these three values

- Given this biological apparatus, we can simulate the presence of many colours by shining Red, Green and Blue light into the human eye with carefully chosen intensities.
- This is the basis on which all colour display technologies (CRTs, LCDs, TFTs, Plasma, Data projectors) operate.
- Inside our machine (TV, Computer, Projector) we represent pixel colours using values for Red, Green and Blue (RGB triples) and the video hardware uses these values to generate the appropriate amount of Red, Green and Blue light.

Image Source:https://en.wikipedia.org/wiki/RGB_color_model#/media/File:Barn_grand_tetons_rgb_separation.jpg



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Additive and Subtractive Color

- Additive color
  - Form a color by adding amounts of three primaries
    - CRTs, projection systems
  - Primaries are Red (R), Green (G), Blue
- Subtractive color
  - Form a color by filtering white light with Cyan (C), Magenta (M), and Yellow (Y) filters
    - Printing

Color Models

- The purpose of a color model is to facilitate the specification of colors in some standard way.
- A color model provides a coordinate system and a subspace in it where each color is represented by a single point.

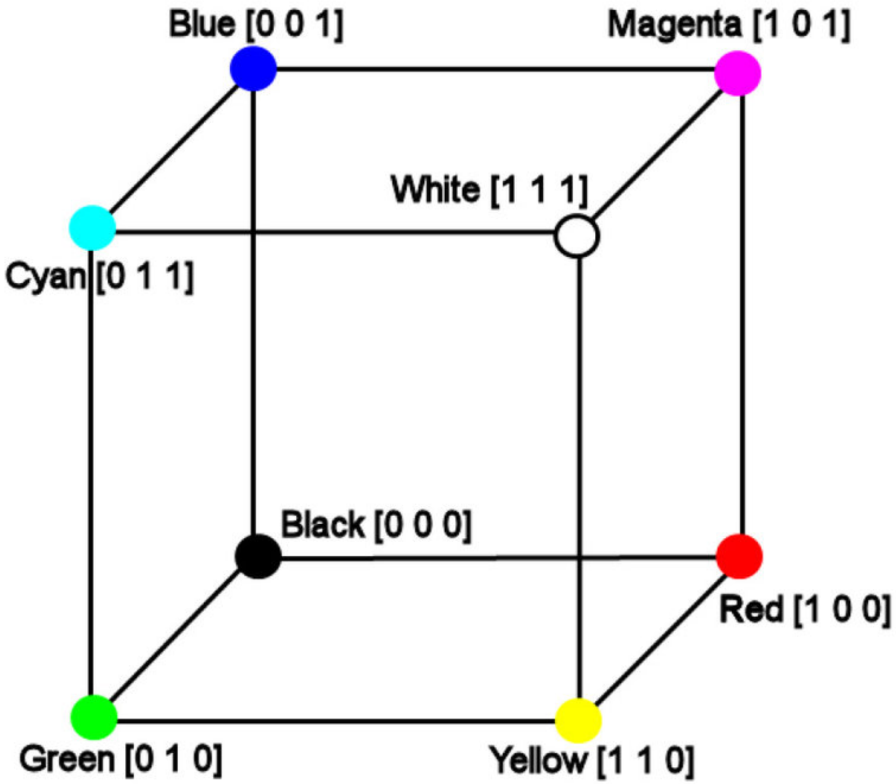
RGB Model

- RGB (Red, Green and Blue) is the color space for digital images.
- Uses three primary colors and create a larger range of colors.
- Used in light emitting devices
  - Color CRT monitors
- Additive
  - Individual contributions of each primary color added together
  - $C = rR + gG + bB$, where $r, g, b \in [0, 1]$

RGB and additive mixing

- A light source within a device creates any color you need by mixing red, green and blue and varying their intensity.
- All colors begin as black darkness. Then red, green and blue light is added on top of each other to brighten it and create the perfect pigment.
- When red, green and blue light is mixed together at equal intensity, they create pure white.

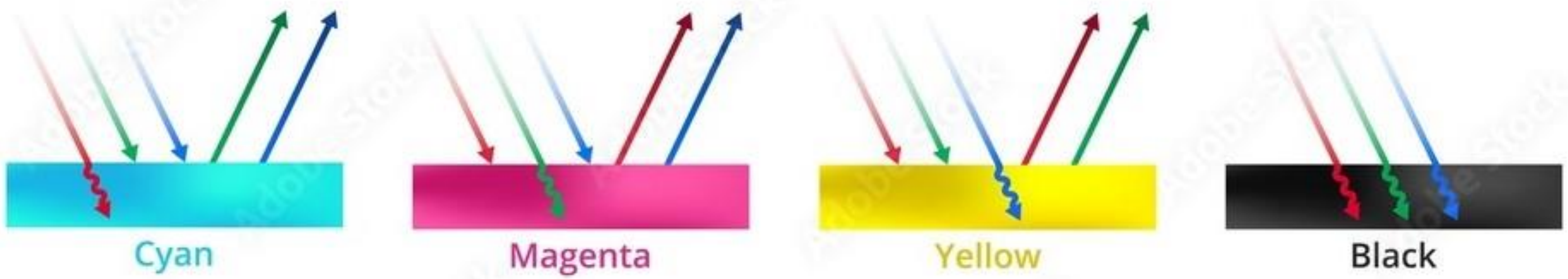
RGB Color Cube



- Color Cube
  - R + G = (1, 0, 0) + (0, 1, 0) = (1, 1, 0) = Y
  - R + B = (1, 0, 0) + (0, 0, 1) = (1, 0, 1) = M
  - B + G = (0, 0, 1) + (0, 1, 0) = (0, 1, 1) = C
  - R + G + B = (1, 1, 1) = W
  - 1 – W = (0, 0, 0) = BLK

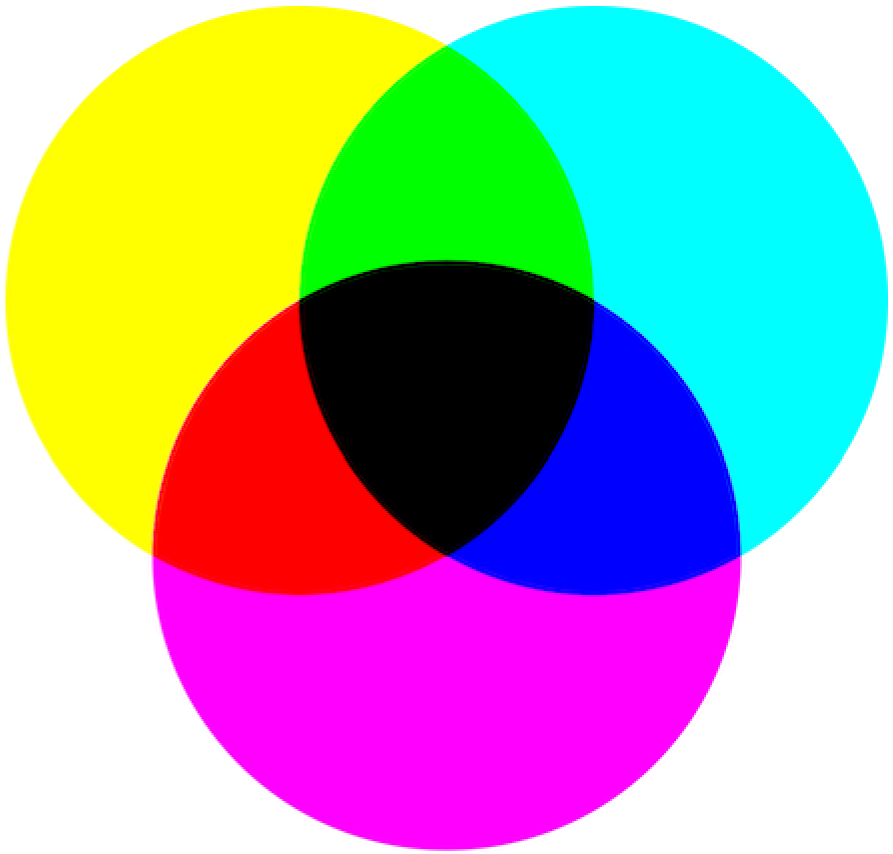
CMY(K) Model

- CMY: Complements of RGB
- Used in light absorbing devices
  - Hardcopy/Printing devices
- Subtractive
  - Color specified by what is subtracted from white light.
  - Cyan absorbs red, magenta absorbs green, and yellow absorbs blue
- Colors are obtained by the reflection of light.



CMY(K) Model Cont.

- C = G+B = W-R
- M = R+B = W-G
- Y = R+G = W-B



CMYK and subtractive mixing

- Subtractive mixing :
  - A printing machine creates images by combining CMYK colors to varying degrees with physical ink.
  - All colors start as blank white, and each layer of ink reduces the initial brightness to create the preferred color.
  - When all colors are mixed together, they create pure black.
  - From where the K comes?
    - According to the theory, 100% cyan, 100% magenta and 100% yellow would result in a pure black.
    - With today's printing colors it is not possible to realize this, so in the area of printing the additional component key (K, black) is necessary.
      - Black ink is cheaper than the coloured ones


CYAN
MAGENTA
YELLOW
BLACK
FINAL CMYK
DETAIL VIEW


2,500 × 1,283

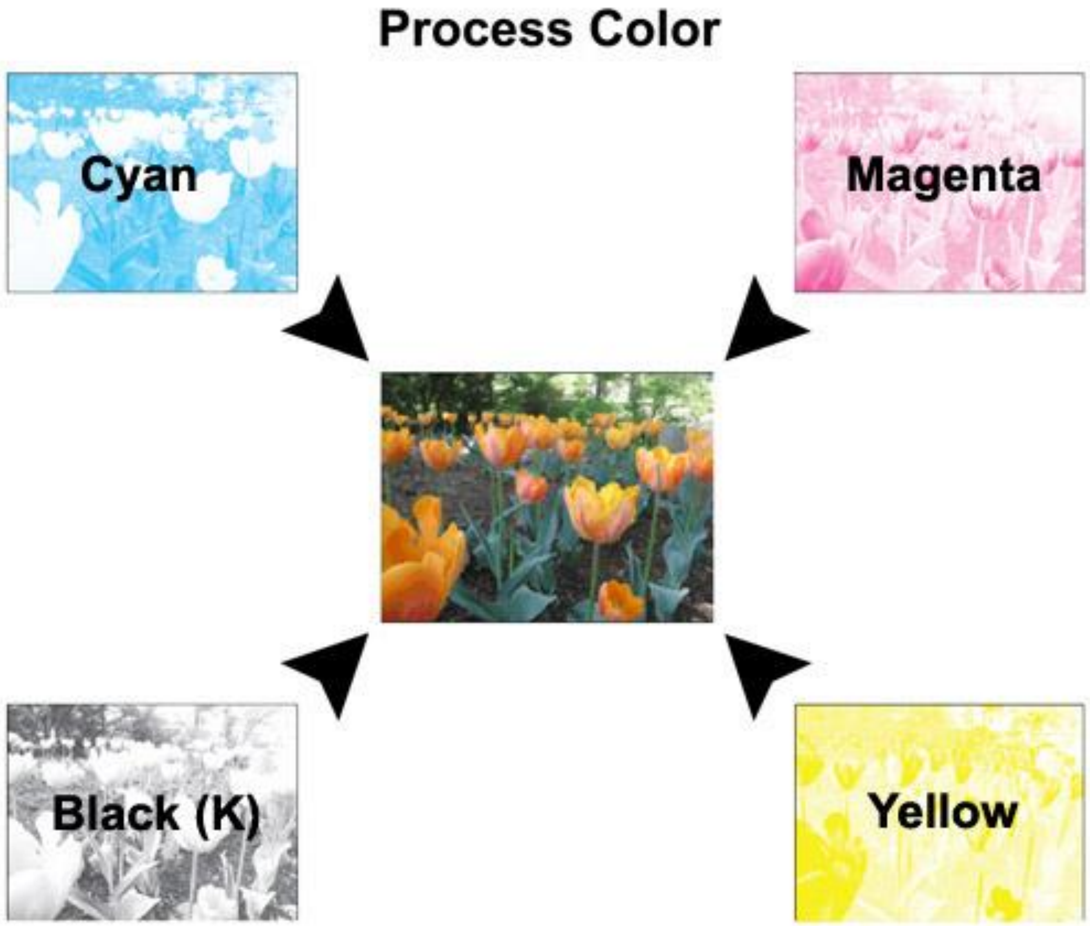

Process Color
Cyan
Magenta
Black (K)
Yellow

Image Source: https://en.m.wikiversity.org/wiki/File:Subtractive-Additive-Colour-Mixing.jpg

https://youtu.be/Rj8sEI1ts3M

Activity 1

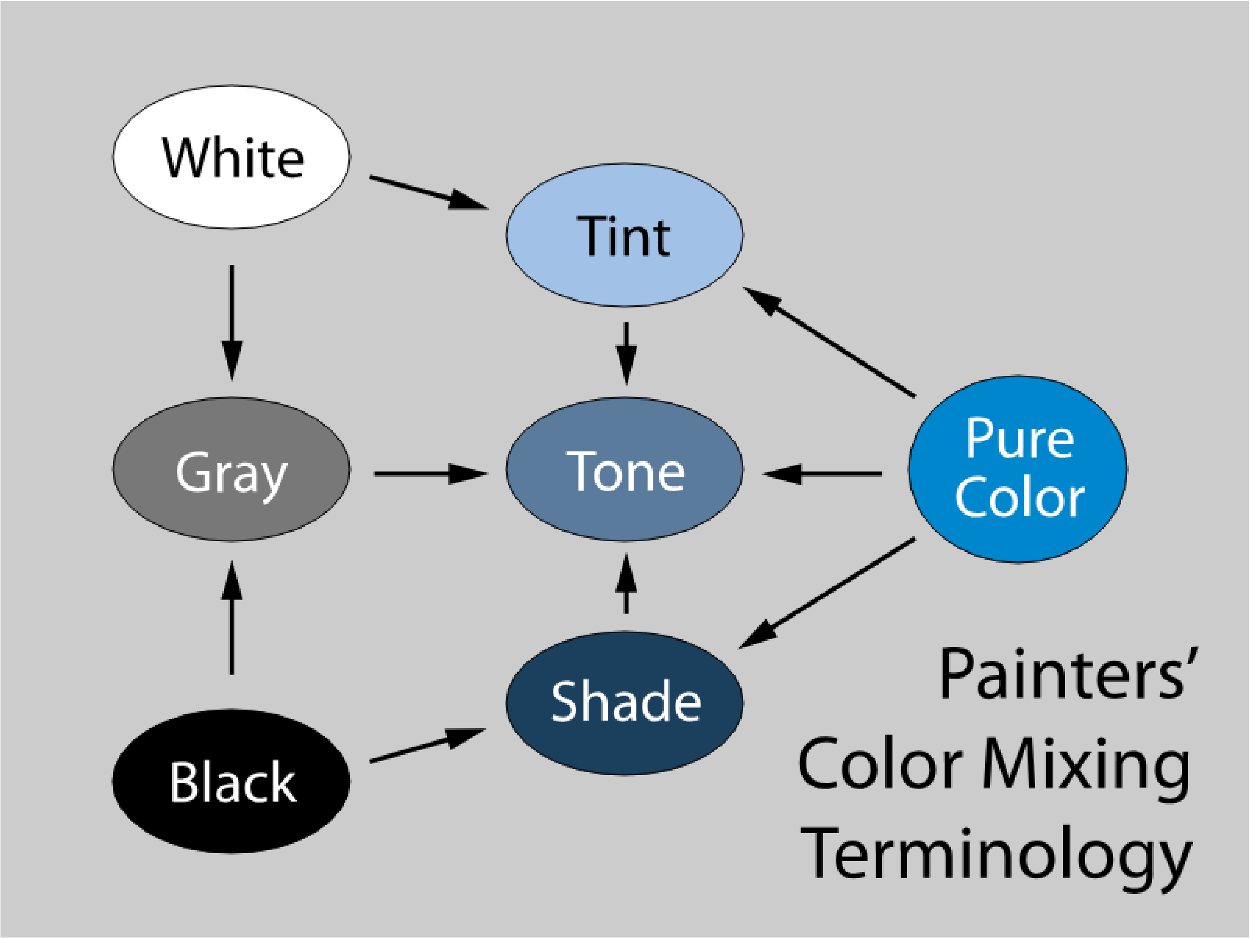
- Briefly explain how to convert between RGB and CMYK.
- Elaborate why the conversion between CMYK and RGB and back again cannot expect identical values.

How Do We Specify Color?

- Color perception usually involves three quantities:
  - Hue: Distinguishes between colors like red, green, blue, etc.
  - Saturation: How far the color is from a gray of equal intensity
  - Lightness: The perceived intensity of a reflecting object
- Sometimes the lightness is called as brightness if the object is emiting light instead of reflecting light.
- In order to use color precisely in computer graphics, we need to be able to specify and measure colors.

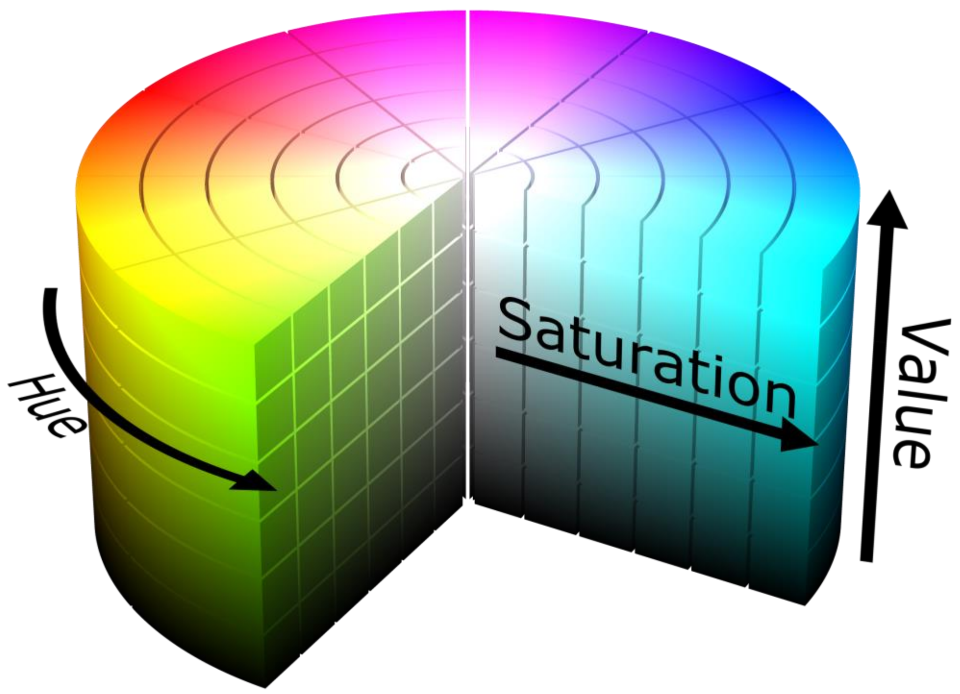
How Do Artists Do It?

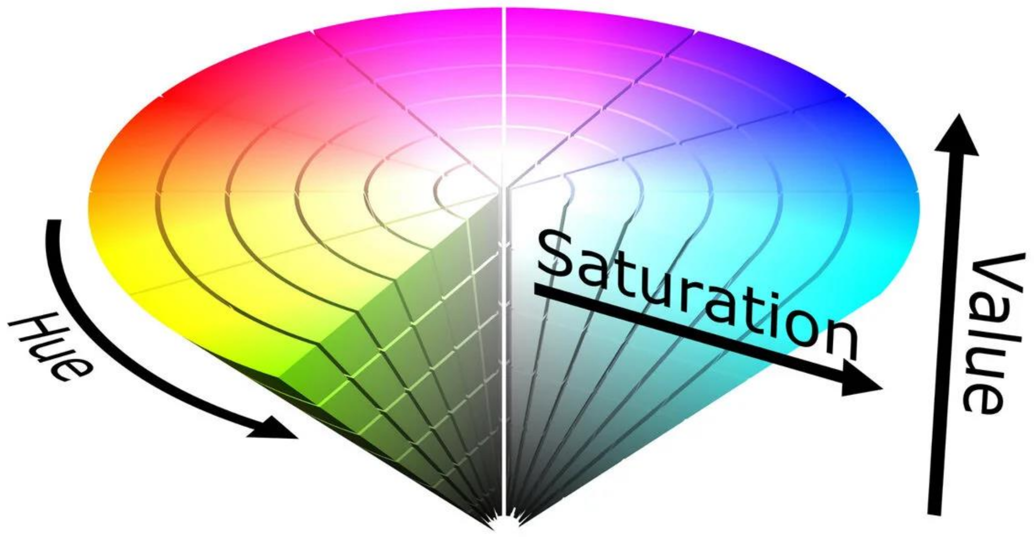
- Artists often specify color as tints, shades, and tones of saturated(pure) pigments.
  - Tint: Adding white to a pure pigment, decreasing saturation
  - Shade: Adding black to a pure pigment, decreasing lightness
  - Tone: Adding white and black to a pure pigment



HSV Color Model

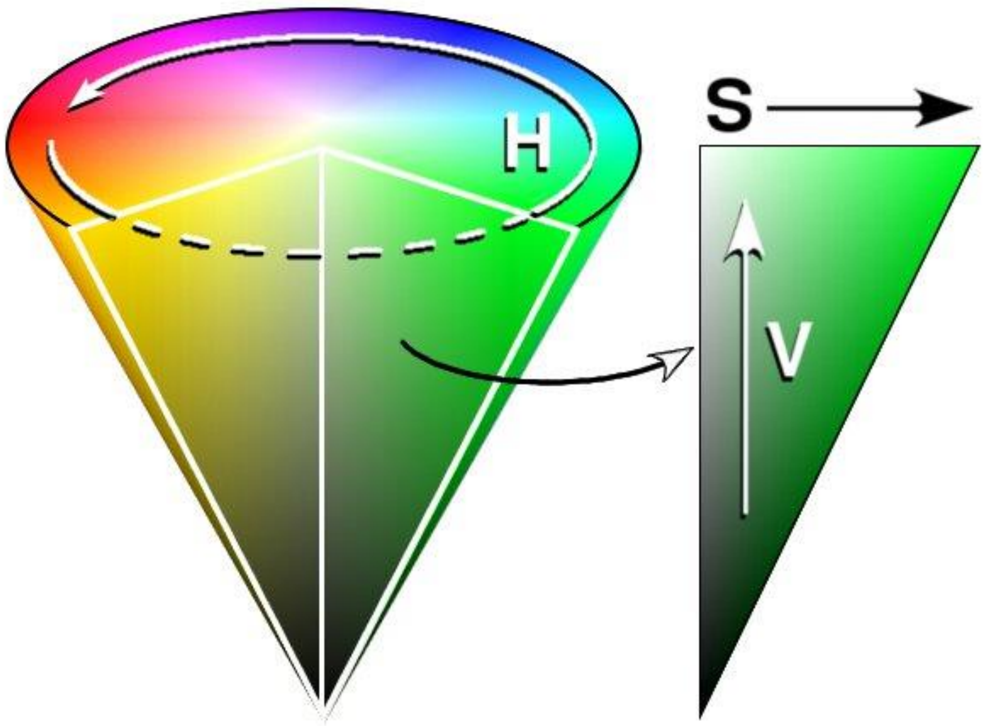
- Alternate way of specifying color
- User-Oriented
- Basis of popular style of color picker





HSV/HSB Color Model-Components

- Hue(H):
  - Hue is the color portion given by angle
  - Expressed as a number from 0 to 360 degrees
    - 0-60 – Red, 61-120 – Yellow, etc.
- Saturation(S):
  - How pure?
  - The amount of gray in a particular color
  - distance from axis
  - 0-gray,1-color(Or 0-gray, 100-color)
- Value(V):
  - Or Brightness(B)
  - Distance along axis
  - 0-black,1-color(Or 0-black, 100-color)



Activity 2

- Briefly explain how to convert between RGB and HSV.

Take Home Assignment

- Prepare a report considering following color models. Briefly explain each of the following models and how these models are used in graphics.
  - RGB
  - CMY(K)
  - YIQ
  - YUV
  - YCbCr
  - HSV
  - HSL
- Include the answers for the Activity 1 and Activity 2 in the report.
- You are required to submit the report by 27^{th} 11.59 PM.
- Important: To avoid being caught for plagiarism, ensure that you include proper references.

Further References

- Chapter 19, Color - Marschner, S., & Shirley, P. (2015). Fundamentals of Computer Graphics. A K Peters/CRC Press. 4th Edition, ISBN-10 : 9781482229394.
- https://www.youtube.com/watch?v=fh7mi9qRXdU
- https://www.youtube.com/watch?v=l7SsfZxy9ms
- https://www.youtube.com/watch?v=gJ2HOj22gDo

Questions?